

CUDDALORE, India

WMO: 433290

Lat: 11.767N

Lon: 79.767E

Elev: 12

StdP: 101.18

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.3	21.0	17.0	12.1	24.7	17.8	12.8	24.7	4.9	27.0	4.1	27.3	0.6	340	0.322

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.6	37.9	27.0	36.9	26.9	36.0	26.8	29.2	33.9	28.8	33.5	28.4	33.0	1.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.4	31.9	27.7	23.6	31.6	27.2	23.0	31.2	96.1	34.2	94.0	33.5	92.1	33.2	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 4.3	3.4	3.1	18.4	40.4	1.1	1.1	17.7	41.2	17.1	41.8	16.5	42.4	15.7	43.2		
(5)			16.6	30.0	1.3	0.7	15.7	30.5	14.9	30.9	14.2	31.3	13.3	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.7	25.4	26.4	28.2	30.3	31.7	31.6	30.7	30.1	29.5	28.3	26.5	25.5		
	DBStd	2.55	1.01	1.13	1.33	1.28	1.68	1.50	1.37	1.29	1.19	1.43	1.21	1.02		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6821	477	459	564	610	672	647	642	622	586	566	495	482		
	CDD18.3	3779	219	226	306	360	414	397	383	364	336	307	245	223		
	CDH23.3	44766	1671	2102	3574	4906	5798	5513	5122	4677	4169	3414	2157	1661		
	CDH26.7	20659	378	652	1476	2553	3361	3151	2712	2318	1913	1289	533	324		
(14) Wind	WSAvg	1.1	1.1	1.0	1.0	1.2	1.4	1.4	1.2	1.1	1.0	0.8	1.0	1.1		
(15) Precipitation	PrecAvg	1384	26	20	13	7	38	37	92	145	141	269	403	201		
	PrecMax	2344	219	347	94	50	381	115	214	366	371	619	988	646		
	PrecMin	733	0	0	0	0	0	4	7	23	3	70	104	5		
	PrecStd	357	49	69	23	11	80	32	56	88	80	153	228	173		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.7	32.4	34.3	37.0	39.9	39.0	38.0	37.1	36.1	34.6	32.1	30.6		
		MCWB	24.8	25.4	26.7	27.8	27.4	26.8	26.4	26.4	25.9	26.3	26.0	25.1		
	2%	DB	30.1	31.4	33.3	35.2	38.1	37.7	36.8	35.8	34.9	33.3	31.0	29.8		
		MCWB	24.1	25.1	26.7	28.0	27.4	26.7	26.3	26.3	26.3	26.5	25.8	24.6		
	5%	DB	29.5	30.6	32.6	34.4	36.8	36.7	35.8	34.9	34.0	32.5	30.2	29.2		
		MCWB	23.8	24.6	26.4	27.9	27.5	26.7	26.2	26.2	26.4	26.4	25.4	24.1		
	10%	DB	28.8	30.0	31.9	33.6	35.5	35.6	34.7	33.9	33.1	31.5	29.5	28.6		
		MCWB	23.4	24.2	26.0	27.6	27.7	26.7	26.1	26.3	26.5	26.4	25.3	23.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.0	26.7	28.2	29.6	30.0	29.5	29.0	28.6	28.2	28.1	27.3	26.6		
		MCDB	29.7	31.0	32.2	33.6	34.4	34.7	34.1	33.7	32.7	31.5	30.1	29.0		
	2%	WB	25.2	26.0	27.6	28.9	29.3	28.8	28.1	28.0	27.7	27.6	26.8	25.9		
		MCDB	28.4	30.0	31.5	33.3	33.9	34.1	33.1	32.4	31.8	31.1	29.6	28.3		
	5%	WB	24.7	25.5	27.2	28.5	28.9	28.2	27.6	27.6	27.4	27.2	26.4	25.4		
		MCDB	27.8	29.3	31.1	32.7	33.6	33.5	32.5	31.9	31.5	30.7	28.9	27.6		
	10%	WB	24.2	25.1	26.7	28.1	28.5	27.7	27.1	27.2	27.1	26.8	26.0	24.9		
		MCDB	27.3	28.5	30.6	32.2	33.1	32.9	32.0	31.5	31.1	30.2	28.4	27.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.0	7.9	7.3	8.6	8.9	8.6	8.3	7.7	6.5	5.7	6.2		
		MCDBR	7.9	8.3	8.2	8.0	10.1	10.1	9.9	9.5	8.9	7.8	7.0	7.1		
	5% WB	MCWBR	3.2	3.1	3.0	2.9	3.8	3.5	3.6	3.6	3.3	3.0	2.9	3.1		
		MCWBR	6.7	7.3	7.4	7.5	8.7	9.4	9.1	8.6	7.9	6.9	5.7	5.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.460	0.453	0.495	0.515	0.594	0.525	0.532	0.522	0.516	0.538	0.508	0.503		
		taud	2.207	2.228	2.114	2.109	1.908	2.122	2.099	2.132	2.149	2.074	2.133	2.105		
	Edn at Noon	Edn at Noon	842	865	834	809	734	777	774	793	804	779	791	790		
		Edh at Noon	143	144	165	164	197	157	161	158	157	166	152	155		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.01	5.91	6.39	6.41	6.07	5.62	5.34	5.45	5.47	4.72	4.13	4.19		
	RadStd	RadStd	0.21	0.23	0.29	0.28	0.39	0.38	0.35	0.28	0.34	0.28	0.45	0.37		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.28	N/A	N/A	+0.62	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page